Test Series No.

**1.3**

# MASTERY TEST SERIES

**DURATION: 90 Minutes** | **DROPPER** | **M. MARKS: 400**

**DATE: 10/02/2026**

## ANSWER KEY

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | (3) | 26. | (2) | 51. | (3) | 76. | (3) |
| 2. | (2) | 27. | (4) | 52. | (1) | 77. | (3) |
| 3. | (4) | 28. | (2) | 53. | (2) | 78. | (2) |
| 4. | (1) | 29. | (1) | 54. | (4) | 79. | (3) |
| 5. | (3) | 30. | (2) | 55. | (1) | 80. | (2) |
| 6. | (3) | 31. | (3) | 56. | (3) | 81. | (2) |
| 7. | (2) | 32. | (1) | 57. | (2) | 82. | (3) |
| 8. | (3) | 33. | (4) | 58. | (2) | 83. | (1) |
| 9. | (3) | 34. | (1) | 59. | (2) | 84. | (1) |
| 10. | (1) | 35. | (4) | 60. | (4) | 85. | (2) |
| 11. | (3) | 36. | (1) | 61. | (3) | 86. | (1) |
| 12. | (4) | 37. | (2) | 62. | (3) | 87. | (1) |
| 13. | (3) | 38. | (4) | 63. | (3) | 88. | (2) |
| 14. | (2) | 39. | (1) | 64. | (2) | 89. | (3) |
| 15. | (1) | 40. | (1) | 65. | (3) | 90. | (2) |
| 16. | (2) | 41. | (3) | 66. | (3) | 91. | (3) |
| 17. | (3) | 42. | (1) | 67. | (4) | 92. | (2) |
| 18. | (1) | 43. | (3) | 68. | (2) | 93. | (2) |
| 19. | (1) | 44. | (3) | 69. | (3) | 94. | (2) |
| 20. | (1) | 45. | (3) | 70. | (2) | 95. | (2) |
| 21. | (2) | 46. | (4) | 71. | (1) | 96. | (1) |
| 22. | (3) | 47. | (4) | 72. | (3) | 97. | (2) |
| 23. | (2) | 48. | (2) | 73. | (4) | 98. | (1) |
| 24. | (4) | 49. | (4) | 74. | (2) | 99. | (3) |
| 25. | (3) | 50. | (4) | 75. | (4) | 100. | (3) |



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**1. (3)**

Time period of pendulum,

$$T_p = 2\pi\sqrt{\frac{\text{string length}}{\text{downward acceleration}}} = 2\pi\sqrt{\frac{L}{a_y}}$$

If the lift is stationary, $a_y = g$.

Time period is $T = 2\pi\sqrt{\frac{L}{g}}$

If the lift is accelerating with $\frac{g}{6}$

Time period $T' = 2\pi\sqrt{\frac{L}{7g/6}}$

$$= \sqrt{\frac{6}{7}}\left(2\pi\sqrt{\frac{L}{g}}\right) = \sqrt{\frac{6}{7}}T$$

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**2. (2)**

$V_{max} = A\omega$

$V_{\omega} = \frac{\omega}{k}$

$$\frac{V_{max}}{V_{\omega}} = \frac{A\omega}{\omega/k} = kA = \frac{2\pi}{\lambda}\times A = \frac{2\pi}{5\pi}\times 10 = 4$$

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**3. (4)**

Process $AB$ is isochoric $\Rightarrow W_1 =$ zero

Process $BC$ is isothermal $\Rightarrow$

$$W_2 = nR(4T_0)\ln\left(\frac{4V_0}{2V_0}\right) = 8RT_0\ln(2)$$

Total work done by gas = $W_1 + W_2 = 8RT_0\ln(2)$

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**4. (1)**

A gas offers no resistance to change in shape, its rigidity is zero.

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**5. (3)**

The wave propagates in negative $x$ direction.

$K = 2 = \frac{2\pi}{\lambda}$

$\therefore\ \lambda = \pi$ m

$A = \frac{1}{10}$ m $= 10$ cm

$\omega = 60 = 2\pi f$

$\therefore\ f = \frac{30}{\pi}$ Hz

Wave velocity $= \frac{\omega}{K} = \frac{60}{2} = 30$ m/s

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**6. (3)**

Equivalent spring constant,

$$k_{eq} = \frac{\left(\frac{k}{3}\right)\left(\frac{2k}{3}\right)}{\frac{k}{3}+\frac{2k}{3}} = \frac{2k}{9}$$

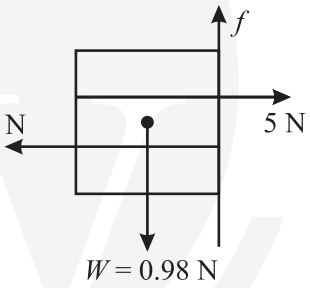
Time period, $T = 2\pi\sqrt{\frac{m}{k_{eq}}} = 2\pi\sqrt{\frac{9m}{2k}} = 6\pi\sqrt{\frac{m}{2k}}$

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**7. (2)**

$N = 5$ N

$f_{max} = \mu N = (0.5)(5) = 2.5$ N




For vertical equilibrium of the block,

$f = mg = 0.98$ N $< (f)_{max}$

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**8. (3)**

$$s_n = u + \frac{a}{2}(2n-1)$$

Since the block starts from rest u = 0

So, $s_n = \frac{a}{2}\times(2n-1)$

For the 5th second (n = 5):

$s_5 = \frac{a}{2}(2\times5-1)$

$s_5 = \frac{a}{2}(9)$

$s_5 = 4.5a$

For the 6th second (n = 6):

$s_6 = \frac{a}{2}(2\times6-1)$



[Contd...

$s_6 = \frac{a}{2}(11)$

$s_6 = 5.5a$

Calculating the ratio:

$$\frac{s_5}{s_6} = \frac{4.5a}{5.5a} = \frac{4.5}{5.5} = \frac{45}{55} = \frac{9}{11}$$

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**9. (3)**

Since power is constant, Work = Power × Time

Work = 4.5 × 4 = 18 Joules

Step 1: Find Final Velocity ($v$)

Using the Work-Energy Theorem (Work = Change in Kinetic Energy):

Work = 1/2 $mv^2$

$18 = 0.5 \times 1 \times v^2$

$v^2 = 36$

$v = 6$ m/s

Step 2: Calculate Distance ($x$)

For an object starting from rest with constant power, the distance is related to final velocity and time by the shortcut formula:

Distance = (2/3) × $v$ × $t$

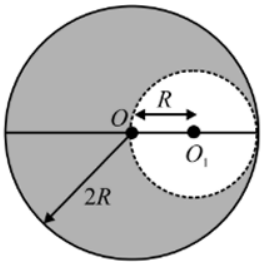
Distance = (2/3) × 6 × 4

Distance = 4 × 4 = 16 m

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**10. (1)**

If mass of bigger disc is $M$ , then mass of removed disc is $\frac{M}{4}$




$$r_{CM} = \frac{M \times 0 - \frac{M}{4} \times R}{M - \frac{M}{4}} = \frac{-\frac{MR}{4}}{\frac{3M}{4}} = -\frac{R}{3}$$

$$\Rightarrow \frac{R}{3} = \alpha R \Rightarrow \alpha = \frac{1}{3}$$

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**11. (3)**

1. Find Angular Velocity

Differentiate the position equation: $\theta(t) = t^{3/3} - 2t^2$

$\omega = d(\theta)/dt = t^2 - 4t$

2. Check Statement I (Reversing Direction):

A particle reverses direction when its velocity becomes zero.

Set $\omega = 0$

$t^2 - 4t = 0$

So, $t = 0$ s (start) and $t = 4$s.

Since the velocity is zero at $t = 4$s, the particle stops momentarily to change direction.

Result: Statement I is Correct.

3. Find Angular Acceleration ($\alpha$):

Differentiate the velocity equation:

$\alpha = d(\omega)/dt = 2t - 4$

4. Check Statement II (Acceleration at Reversal):

We know the reversal happens at $t = 4$s. Let's put this time into the acceleration equation.

$\alpha = 2(4) - 4$

$\alpha = 8 - 4 = 4$ rad/s$^2$.

The acceleration is 4, not zero.

Result: Statement II is Incorrect

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**12. (4)**

$$g = \frac{4}{3}\pi G\rho R$$

$$g_A : g_B : g_C$$

$$= \frac{3}{4}\pi G\rho R : \frac{3}{4}\pi G \times \frac{\rho}{2} \times \frac{3}{2}R : \frac{3}{4}\pi G\frac{\rho}{4} \times 2R$$

$$= 4 : 3 : 2$$

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**13. (3)**

Analyze statement I (calculation):

For fluid to flow into the vein, the pressure exerted by the fluid column must be at least equal to the vein's pressure.

Formula: Pressure = density × g × height

2000 = 1000 × 10 × h

2000 = 10000 × h

h = 2000 / 10000

h = 0.2 meters.

So, Statement I is Correct.

2. Analyze Statement II:

The pressure generated by the fluid column is directly proportional to density (P = ρ × g × h)

If we increase the density (ρ), the Pressure (P) exerted by the fluid column increases.

Since the pressure becomes higher than 2000 Pa, the fluid will flow into the vein even more easily. It will not stop.

So, Statement II is Incorrect.

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**14. (2)**

The value recorded is 5.20 cm, which has two decimal places. This means the instrument must have a least count (LC) of 0.01 cm, which is equal to 0.1 mm.

Let us calculate the least count for each option:

(1) Smallest division is 1 mm, so LC = 0.1 cm. Reading would be like 5.2 cm.

(2) 1 MSD = 1 mm. LC = 1 mm / 10 = 0.1 mm = 0.01 cm. This matches.

(3) LC = Pitch / Divisions = 1 mm / 100 = 0.01 mm = 0.001 cm.

(4) LC = 0.5 mm / 50 = 0.01 mm = 0.001 cm.

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**15. (1)**

$$\rho = \frac{M}{V} \Rightarrow \frac{\Delta\rho}{\rho} = -\frac{\Delta V}{V} = \frac{\Delta P}{B}$$

$$\frac{0.3}{100} = \frac{\Delta P}{2\times10^{11}}$$

$$\Rightarrow \Delta P = 6\times10^{8}\ \text{N/m}^2$$

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**16. (2)**

$$\frac{\text{Reading} - \text{LFP}}{\text{UFP} - \text{LFP}} = \text{constant}$$

$$\frac{C-0}{100-0} = \frac{50-(-10)}{190-(-10)} = \frac{60}{200} = \frac{3}{10}$$

$$C = 100\times\frac{3}{10} = 30°\text{C}$$

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**17. (3)**

$$v_{\text{rms}} = v_{\text{mp}}\sqrt{\frac{3}{2}}$$

$$v_{\text{rms}} = 200\sqrt{\frac{3}{2}}$$

$$v_{\text{rms}} = 200(\sqrt{3}/\sqrt{2})$$

$$v_{\text{rms}} = 100\sqrt{6}\ \text{m/s}$$

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**18. (1)**

$$k_{\text{eff}} = k_1 + k_2$$

$$k_{\text{eff}} = 100 + 300 = 400\ \text{N/m}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{\text{eff}}}{m}} = \frac{1}{2\pi}\sqrt{\frac{400}{4}} = \frac{10}{2\pi} = \frac{5}{\pi}\ \text{Hz}$$

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**19. (1)**

On comparing $\omega = 4\pi = \frac{2\pi}{T}$

$\therefore$ T = 0.5 s

$$v_{\max} = \omega A = 4\pi\times2 = 8\pi\ \text{m/s}$$

$$a_{\max} = \omega^2 A = 16\pi^2\times2 = 32\pi^2\ \text{m/s}^2$$

$$k = m\omega^2 = 1\times16\pi^2 = 16\pi^2\ \text{N/m}$$

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**20. (1)**

Breaking condition: $\sigma_b = \frac{F}{A} \Rightarrow \text{A} = \frac{F}{\sigma_b}$

$$F = mg = 900\times10 = 9\times10^3\ \text{N}$$

$$\Rightarrow \text{A} = \frac{9\times10^3}{4.5\times10^8} = 2\times10^{-5}\ \text{m}^2$$

$$A = \frac{\pi d^2}{4} \Rightarrow \text{d} = \sqrt{\frac{4\text{A}}{\pi}} = \sqrt{\frac{4\left(2\times10^{-5}\right)}{\pi}}$$

$$\approx 5.0\times10^{-3}\ \text{m} = 0.5\ \text{cm}$$

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**21. (2)**

$$\frac{\Delta L}{L_0} = \alpha\Delta T \qquad \ldots(1)$$

$$\frac{\Delta A}{A} = \beta\Delta T \qquad \ldots(2)$$

From (1) and (2)

$$\frac{\Delta A}{A} = 2\frac{\Delta L}{L_0}$$

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**22. (3)**

In closed organ pipe sound wave has a node at the closed end and antinode at the open end $\Rightarrow l = \frac{\lambda}{4}$.

$\therefore$ Fundamental frequency is $f_0 = \frac{v}{\lambda} = \frac{v}{4l}$.

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**23. (2)**

$$\Delta h = H - h = 6 - 1.05 = 4.95\ \text{m}$$

r = 0.1

$$v^2 = \frac{2g\Delta h}{1-r^2} = \frac{2\times10\times4.95}{1-(0.1)^2}$$

$$= \frac{99}{0.99} = 100\ \text{m}^2/\text{s}^2$$

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**24. (4)**

$\Delta U = nC_v\Delta T$ where, $C_v = \dfrac{5R}{2}$

(i) Change in internal energy $A \to B$:

$\Delta U_{AB} = nC_v(T_B - T_A)$

$= 1 \times \dfrac{5R}{2}(800 - 400) = 1000R$

(ii) Change in internal energy $B \to C$:

$\Delta U_{BC} = nC_v(T_C - T_B)$

$= 1 \times \dfrac{5R}{2}(600 - 800) = -500R$

(iii) Change in internal energy $C \to A$:

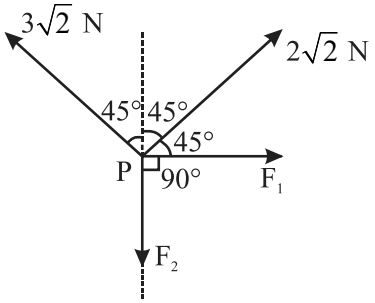
$\Delta U_{CA} = nC_v(T_A - T_C)$

$= 1 \times \dfrac{5R}{2}(400 - 600) = -500R$

(iv) $\Delta U_{\text{Total}} = 0$, in a cyclic process.

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**25. (3)**




$F_1 + 2\sqrt{2} \times \dfrac{1}{\sqrt{2}} = 3\sqrt{2} \times \dfrac{1}{\sqrt{2}}$

$\Rightarrow F_1 = 1\ \text{N}$

$F_2 = 2\sqrt{2} \times \dfrac{1}{\sqrt{2}} + 3\sqrt{2} \times \dfrac{1}{\sqrt{2}} = 2 + 3 = 5\ \text{N}$

Ratio $F_1 : F_2 = 1 : 5$.

Comparing with 3 : $x$:

$\dfrac{3}{x} = \dfrac{1}{5} \Rightarrow x = 15$

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**26. (2)**

Due to –M & –I of groups

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**27. (4)**

Angular momentum in two successive orbitals are

$\frac{nh}{2\pi}$ and $(n+1)\frac{h}{2\pi}$

$\therefore a = \frac{n+1}{n}$ and $b = \frac{h}{2\pi}$

$\therefore \frac{a}{b} = \frac{n+1}{n} \cdot \frac{2\pi}{h}$

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**28. (2)**

A: False. Acetylene is colourless but not strictly odourless.

B: True. Acetylene (ethyne) is slightly more acidic than propyne.

C: False. Baeyer's reagent is cold dilute alkaline $KMnO_4$.

D: True. Ozonolysis of But-2-yne ($CH_3–C\equiv C–CH_3$) gives two molecules of acetic acid.

E: False. Bromine water is decolourised by both alkenes and alkynes

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**29. (1)**

In paper chromatography, a special quality paper known as chromatography paper is used. Chromatography paper contains water trapped in it, which acts as the stationary phase, not the paper material itself.

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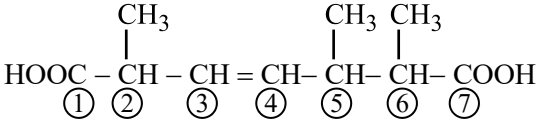
**30. (2)**

| Group-14 | Electronegative value |
|---|---|
| Carbon | 2.5 |
| Silicon | 1.8 |
| Germanium | 1.8 |
| Tin | 1.8 |
| Lead | 1.9 |

The electronegativity values for elements from Silicon to Lead are almost same. Hence Statement I is false.

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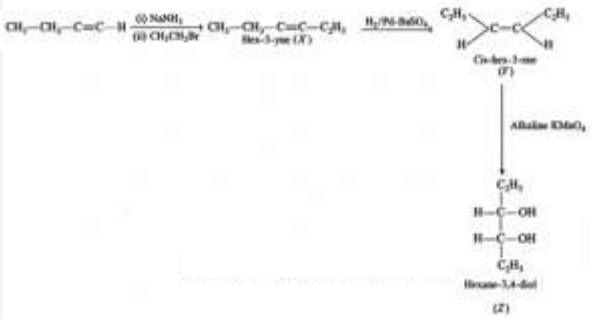
**31. (3)**



2, 5, 6-Trimethyl-3-heptanedioic acid

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**32. (1)**



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**33. (4)**

| | A(aq.) | ⇌ | 2B(aq.) | + | C(aq.) |
|---|---|---|---|---|---|
| mol fraction at t = 0 | $\frac{2}{3}$ | | $\frac{1}{3}$ | | 0 |
| molarity t = 0 | 2M | | 1M | | 0 |
| At eq. | (2M – x) | | (M + 2x) | | x |

Since mole fraction are in the ratio 2 : 1 for A and B thus their molarity will also show same ratio)

Thus total mole at equilibrium

$= 2M - x + M + 2x + x = 3M + 2x$

Given mole fraction of A at eq. $= \frac{1}{2} = \frac{2M-x}{3M+2x}$

$\therefore x = \frac{M}{4}$;

Thus $[B] = M + \frac{2M}{4} = \frac{3M}{2}$

and $[A] = 2M - \frac{M}{4} = \frac{7M}{4}$

Now, $K_C = \frac{[B]^2[C]}{[A]} = \frac{\left[\frac{3M}{2}\right]^2\left[\frac{M}{4}\right]}{\frac{7M}{4}} = \frac{9M^2}{28}$

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**34. (1)**

The boiling point of cis 1,2-Dichloro ethene is higher than that of corresponding trans isomer because 1,2-Dichloro ethene has higher dipole moment as compared to that of the trans isomers.

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**35. (4)**

The technique used is steam distillation for those liquids which are immiscible, containing non-volatile impurities and are steam volatile.

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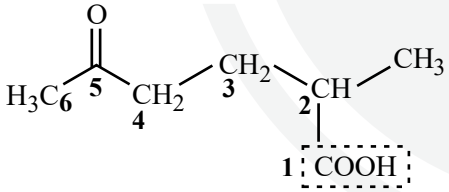
**36. (1)**

Ionisation enthalpy increases along the period but decreases down the group.

**(NCERT Ed. 2025-26 11th Part-I Page No. 87)**

**37. (2)**

In the given compound, the numbering will start from –COOH group. Double bond O present at 5th position will named as 'oxo' in prefix.



IUPAC name of the above compound is:

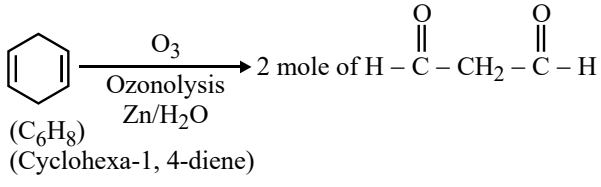
2-Methyl-5-oxohexanoic acid.

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**38. (4)**

Hydrocarbon 'X' is cyclohexa-1,4-diene

The reaction is given below:




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**39. (1)**

For a compound to be aromatic, it must follow Huckel's rule.

In (1), $4n + 2 = 6 \Rightarrow n = 1$ (integer), hence aromatic.

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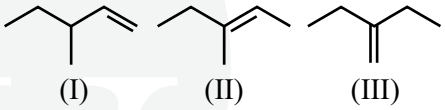
**40. (1)**

| List-I | | List-II | |
|---|---|---|---|
| (A) | $XeO_3$ | (III) | Trigonal pyramid |
| (B) | $XeOF_4$ | (IV) | Square pyramid |
| (C) | $BO_3^{3-}$ | (I) | Planar triangular |
| (D) | $ClF_3$ | (II) | T-shape |

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**41. (3)**

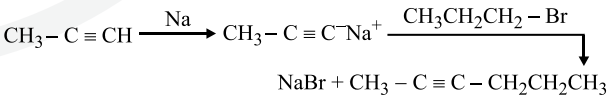
From molar mass, it can be $C_6H_{12}$




(I) has chiral carbon and (II) show cis/trans

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**42. (1)**




**(NCERT Ed. 2025-26 11th Part-II Page No. 315)**

**43. (3)**

In chromatography, the retardation factor ($R_f$) is defined as:

$$R_f = \frac{\text{Distance travelled by the substance (spot)}}{\text{Distance travelled by the solvent front}}$$

From the figure:

x = distance travelled by the spot.

y = distance travelled by the solvent front.

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**44. (3)**

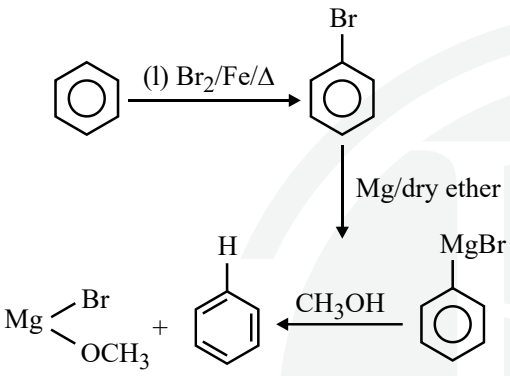
Structure has two terminal Br atoms each with three Br = O double bonds and a central Br with two Br = O double bonds → total π bonds = 3 + 3 + 2 = 8.

$Fe_3O_4$ contains one $Fe^{2+}$ and two $Fe^{3+}$; oxidation to $Fe_2O_3$ (all $Fe^{3+}$) requires the single $Fe^{2+}$ to lose $1e^-$.

Cl atoms are in –1 ($Cl^-$) and +1 (in $OCl^-$) → difference = (+1) – (–1) = 2.

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**45. (3)**




Hence, option (3) is the correct answer.

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**46. (4)**

More electronegative atom having –ve charge and more electropositive having +ve charge.

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**47. (4)**

$$n = \frac{\text{Weight}}{\text{Molar mass}}$$

16 g $CH_4$ = 1 mole $CH_4$

1 mole $CH_4 \rightarrow 10 \times 6.02 \times 10^{23}$ electrons

$= 60.2 \times 10^{23}$ electrons

1 g $H_2$ = 0.5 mole $H_2$

1 mole $H_2$ at STP → 22.4 L

0.5 mole $H_2$ at STP → 11.2 L ≃ 11.4 L

1 mole of $N_2$ = 28 g

($\because M_{N_2} = 28$ g / mol )

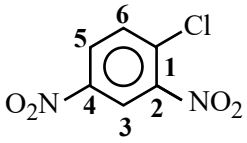
1 mole of $SO_2$ = 64 g

($\because M_{SO_2} = 64$ g / mol )

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**48. (2)**

Statement I:

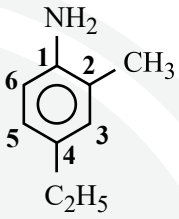



IUPAC name

⇒1-Chloro-2, 4-dinitrobenzene

Hence, statement-I is incorrect

Statement II:




IUPAC name

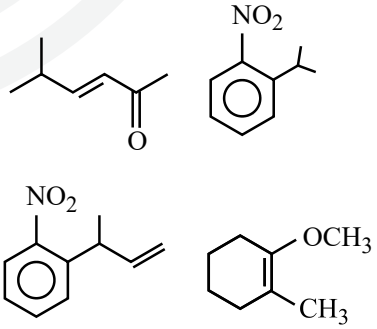
⇒4-Ethyl-2-methylaniline

Hence, statement-II is correct

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**49. (4)**

The compounds that shows the given mentioned effects are:-




Hence, in total we have four compounds.

**(NCERT Ed. 2025-26 11th Part-II Page No. 277)**

**50. (4)**

Ratio of slope $= \frac{\gamma \text{ for gas I}}{\gamma \text{ for gas II}}$; slope for He, $N_2$ and $O_3$ are 5/3, 7/5 and 4/3 respectively.

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[Contd...

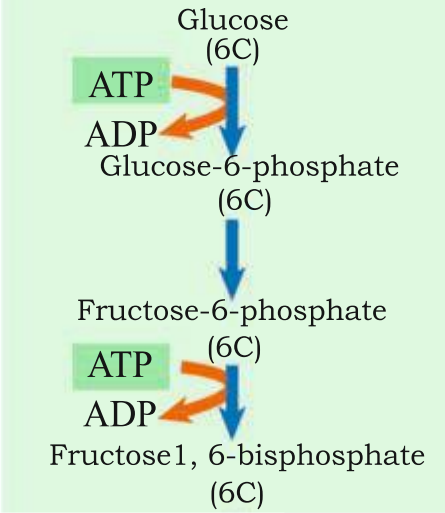
**51. (3)**

| | |
|---|---|
| Telophase | Nucleolus, Golgi complex and ER reform |
| Prophase | Initiation of condensation of chromosomal material |
| Metaphase | Spindle fibres attach to the kinetochores of chromosomes |
| Anaphase | Centromeres split and chromatids separate |

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**52. (1)**

A-ATP and B-ATP




**Steps of glycolysis**

**(NCERT Ed. 2025-26, 11th, Page No. 156)**

**53. (2)**

In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells.

**(NCERT Ed. 2025-26, 11th, Page No. 77)**

**54. (4)**

When fats are used in respiration, the RQ is less than 1. When fats are used in respiration, the RQ is less than 1.

**(NCERT Ed. 2025-26, 11th, Page No. 164)**

**55. (1)**

The number of species that are known and described ranges between 1.7-1.8 million.

**(NCERT Ed. 2025-26, 11th, Page No. 3)**

**56. (3)**

The NADP reductase enzyme is located on the stroma side of the membrane.

**(NCERT Ed. 2025-26, 11th, Page No. 141)**

**57. (2)**

When cells photosynthesise, the availability of $O_2$ is not a problem in these cells since $O_2$ is released within the cell.

**(NCERT Ed. 2025-26, 11th, Page No. 154)**

**58. (2)**

The fixation of six molecules of $CO_2$ and 6 turns of the cycle are required for the formation of one molecule of glucose from the pathway. For two molecules of glucose 12 $CO_2$ molecules and 12 turns are required.

**(NCERT Ed. 2025-26, 11th, Page No. 144)**

**59. (2)**

In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid.

NADH is oxidised to $NAD^+$ rather slowly in fermentation.

**(NCERT Ed. 2025-26, 11th, Page No. 162)**

**60. (4)**

In prokaryotes, ribosomes are made of two subunits -50S and 30S units which when present together form 70S prokaryotic ribosomes. Ribosomes are the site of protein synthesis. Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome.

**(NCERT Ed. 2025-26, 11th, Page No. 91)**

**61. (3)**

The $C_4$ plants show saturation at about 360 $\mu l L^{-1}$ while $C_3$ responds to increased $CO_2$ concentration and saturation is seen only beyond 450 $\mu l L^{-1}$.

**(NCERT Ed. 2025-26, 11th, Page No. 150)**

**62. (3)**

Glucose and fructose are phosphorylated to give rise to glucose-6- phosphate by the activity of the enzyme hexokinase.

**(NCERT Ed. 2025-26, 11th, Page No. 156)**

**63. (3)**

| | |
|---|---|
| Diatoms | The cell wall forms two thin overlapping shells |
| *Nostoc* | Gelatinous sheath |
| Dinoflagellates | Stiff cellulose plates |
| Euglenoids | Pellicle |

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[Contd...

**64. (2)**

Complex IV refers to cytochrome *c* oxidase complex containing cytochromes *a* and $a_3$ , and two copper centres.

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**65. (3)**

Chlorophyll *a* (bright or blue green in the chromatogram), chlorophyll *b* (yellow green), xanthophylls (yellow) and carotenoids (yellow to yellow-orange).

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**66. (3)**

Glycerol would enter the pathway after being converted to PGAL.

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**67. (4)**

The majority of the red algae are marine.
The red thalli of most of the red algae are multicellular.
Sexual reproduction is oogamous.
Agar is obtained from members of red algae.
Carrageen is obtained from red algae.

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**68. (2)**

$CF_0$ is embedded in the thylakoid membrane and forms a transmembrane channel that carries out facilitated diffusion of protons across the membrane.

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**69. (3)**

Carbon dioxide is the major limiting factor for photosynthesis.

**(NCERT Ed. 2025-26, 11th, Page No. 150)**

**70. (2)**

In yeast, the anaerobic breakdown of pyruvic acid results in the formation of ethanol and $CO_2$.

**(NCERT Ed. 2025-26, 11th, Page No. 157)**

**71. (1)**

The splitting of water is associated with the PS II; water is split into $2H^+$, [O] and electrons. This creates oxygen, one of the net products of photosynthesis.

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**72. (3)**

Initial fixation of $CO_2$ occurs in mesophyll cells to form OAA, which is then converted to malic acid or aspartic acid in the mesophyll cells of $C_4$ plants.

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**73. (4)**

During the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised. This is a substrate level phosphorylation. In a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP. There are three points in the cycle where $NAD^+$ is reduced to NADH + $H^+$ and one point where $FAD^+$ is reduced to $FADH_2$. The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of oxaloacetic acid, the first member of the cycle.

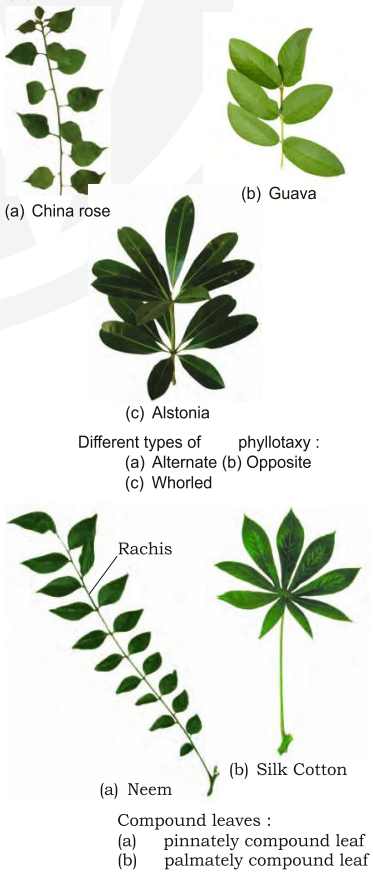
**(NCERT Ed. 2025-26, 11th, Page No. 159)**

**74. (2)**

During glycolysis 3-phosphoglyceraldehyde (PGAL) is converted to 1, 3-bisphosphoglycerate (BPGA). Two redox-equivalents are removed (in the form of two hydrogen atoms) from PGAL and transferred to a molecule of $NAD^+$. PGAL is 6 oxidized and with inorganic phosphate to get converted into BPGA. The conversion of BPGA to 3-phosphoglyceric acid (PGA), is also an energy yielding process; this energy is trapped by the formation of ATP.

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**75. (4)**




**(NCERT Ed. 2025-26, 11th, Page No. 61)**

**76. (3)**

Macrophages and leucocytes exhibit amoeboid movement, which involves the streaming of protoplasm and cytoskeletal elements like microfilaments. Ciliary movement is used for the passage of ova and removing dust from the trachea, while limbs require muscular movement.

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**77. (3)**

The Peripheral Nervous System (PNS) is divided into the somatic system (CNS to skeletal/voluntary muscles) and the autonomic system (CNS to involuntary organs/smooth muscles).

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**78. (2)**

Each meromyosin has a globular head with a short arm (HMM) and a tail (LMM). The HMM component projects outwards at a regular distance and angle from the surface of the filament and is known as the cross arm.

**(NCERT Ed. 2025-26, 11th, Page No. 221)**

**79. (3)**

In a resting neuron, the membrane is more permeable to $K^+$ and nearly impermeable to $Na^+$. It is also impermeable to the negatively charged proteins inside, which helps maintain the negative resting potential inside the axon.

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**80. (2)**

| List-I | List-II |
|---|---|
| Fibrous joint | Between flat skull bones |
| Cartilaginous joint | Between adjacent vertebrae |
| Ball and Socket joint | Between humerus and pectoral girdle |
| Gliding joint | Between carpals |

**(NCERT Ed. 2025-26, 11th, Page No. 227)**

**81. (2)**

The thalamus serves as the major coordinating and relay center for sensory and motor signals to the cerebral cortex (except olfactory impulses).

**(NCERT Ed. 2025-26, 11th, Page No. 236)**

**82. (3)**

Serum is defined as plasma without clotting factors.

Basophils are the least abundant WBCs and are not phagocytic.

The most abundant WBCs are neutrophils, which are phagocytic.

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**83. (1)**

In a resting state, a complex protein called troponin is distributed at regular intervals on the tropomyosin. A specific subunit of this troponin masks the active binding sites for myosin on the actin filaments to prevent contraction without a signal.

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**84. (1)**

An increase in body fluid volume suppresses ADH release. ADH is released when body fluid volume decreases or osmolarity increases.

Angiotensin II is a powerful vasoconstrictor, and it increases glomerular blood pressure.

A fall in GFR stimulates juxtaglomerular (JG) cells to release renin.

Atrial Natriuretic Factor (ANF) antagonizes the renin–angiotensin–aldosterone system, helping reduce blood volume and pressure.

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**85. (2)**

Bipolar neurons occur in the retina, olfactory epithelium, and inner ear.

Unipolar neurons are mainly present during the embryonic stage; in adults they usually become pseudounipolar.

Multipolar neurons have one axon and many dendrites, not just one dendrite.

Both spinal and cranial nerves contain myelinated fibers for rapid impulse conduction.

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**86. (1)**

Myasthenia gravis is an autoimmune disorder where the body's immune system attacks the neuromuscular junction. This prevents signals from reaching the muscle, leading to fatigue, weakening, and eventually paralysis.

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**87. (1)**

The duration of one cardiac cycle in a normal adult at rest is about 0.8 seconds.

During joint diastole, all four chambers are relaxed, but the semilunar valves are closed, not open.

Atrial systole contributes about 20–30% additional blood to the ventricles.

The second heart sound (dub) is due to the closure of the semilunar valves (aortic and pulmonary).

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[Contd...

**88. (2)**

The given figure is of muscles. An organized muscles is held by fascia (connective tissue). Inside are muscle bundles (fascicles). Each bundle contains muscle fibres, and each fibre is lined by the sarcolemma (plasma membrane).

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**89. (3)**

| List-I | List-II |
|---|---|
| Hypothalamus | Controls body temperature and urge for eating |
| Cerebellum | Convoluted surface to provide space for more neurons |
| Medulla | Controls respiration and gastric secretions |
| Corpus callosum | Connects two cerebral hemispheres |

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**90. (2)**

Muscle contraction is initiated by a signal from the CNS via a motor neuron.

During contraction, the I-band shortens, while the A-band remains constant in length.

The H-zone is the central region of the thick filament that is not overlapped by thin filaments (it narrows or disappears during contraction).

Shortening of the sarcomere occurs as the Z-lines are pulled towards each other.

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**91. (3)**

Transmission across an electrical synapse is faster because ions pass directly between neurons through gap junctions.

Chemical synapses require the release, diffusion, and binding of neurotransmitters across the synaptic cleft.

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**92. (2)**

| List-I | List-II |
|---|---|
| Fibrinogen | Coagulation of blood |
| Globulins | Involved in defense mechanisms |
| Albumins | Osmotic balance |
| Spleen | Graveyard of RBCs |

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**93. (2)**

Hydra has a simple network of neurons. Insects show advancement with a brain and ganglia. Vertebrates have the most developed system. The neural system (not endocrine) provides point-to-point quick coordination.

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**94. (2)**

White fibers possess very little myoglobin and fewer mitochondria, making them appear pale. However, they have a high amount of sarcoplasmic reticulum and rely on anaerobic respiration for energy, unlike Red fibers which are aerobic.

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**95. (2)**

Joints are necessary for movements involving bones. Without joints, bones would be rigid and movement would not be possible.

In the musculoskeletal system, muscles provide the force (effort), bones act as levers, and joints act as the fulcrum. The load is usually the weight of the body part or an external object being moved.

**(NCERT Ed. 2025-26, 11th, Page No. 227)**

**96. (1)**

| List-I | List-II |
|---|---|
| Protonephridia | Flatworms |
| Nephridia | Earthworms |
| Malpighian tubules | Cockroaches |
| Antennal glands | Prawns |

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**97. (2)**

Descending limb of Loop of Henle is highly permeable to water and almost impermeable to electrolytes, allowing water to move out and concentrate the filtrate.

Ascending limb of Loop of Henle is impermeable to water but permeable to electrolytes ($Na^+$, $Cl^-$), with active transport in the thick ascending limb.

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**98. (1)**

The cerebral cortex has association areas that integrate sensory and motor information and are not exclusively sensory or motor.

The midbrain lies between the thalamus (forebrain) and the pons (hindbrain), serving as a connecting region.

**(NCERT Ed. 2025-26, 11th, Page No. 236)**

**99. (3)**

During the first delivery, the mother is exposed to $Rh^{+ve}$ blood and prepares antibodies. In subsequent pregnancies, these antibodies cross the placenta and destroy the foetal RBCs, causing severe anaemia and jaundice.

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**100. (3)**

Humans have 12 pairs of ribs. The first 7 are true (vertebrosternal). The 8th, 9th, and 10th join the 7th rib (vertebrochondral/false). The last 2 pairs (11th and 12th) are not connected ventrally, hence they are called floating ribs. Ribs are bicephalic (two articulation surfaces).

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